

BridgeForge

Detect contract drift.
Synchronize dependent artifacts
automatically.

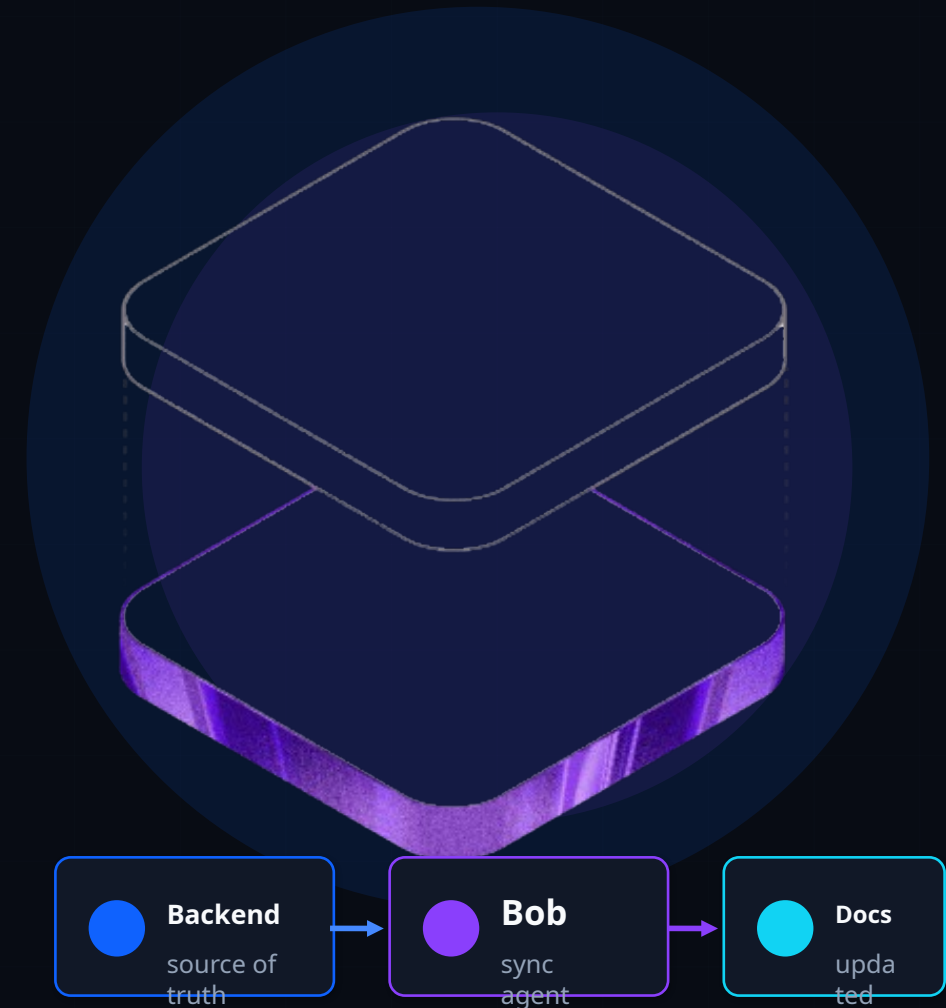
An AI-powered API contract synchronization layer built around
IBM Bob Shell.

Node.js / Express

Chokidar Watcher

IBM Bob Shell

React Dashboard



Cross-stack auto-sync prototype

API contract drift hides inside normal refactors

A one-field backend change can instantly make downstream artifacts disagree with the source of truth.

A “small” backend change

products.js

```
const {  
  title, price } = req.body;  
  ↓ rename  
const {  
  productName, price } = req.body;
```

The backend is correct immediately. Everything that still expects `title` is now at risk.

Hidden coupling

The hard part is not changing the API — it is finding every dependent artifact.



API docs go stale

Human-readable examples and tables still describe the old field.



OpenAPI diverges

Machine-readable schemas no longer match the implementation.



Clients can mismatch

Frontend code may still bind to the previous response shape.



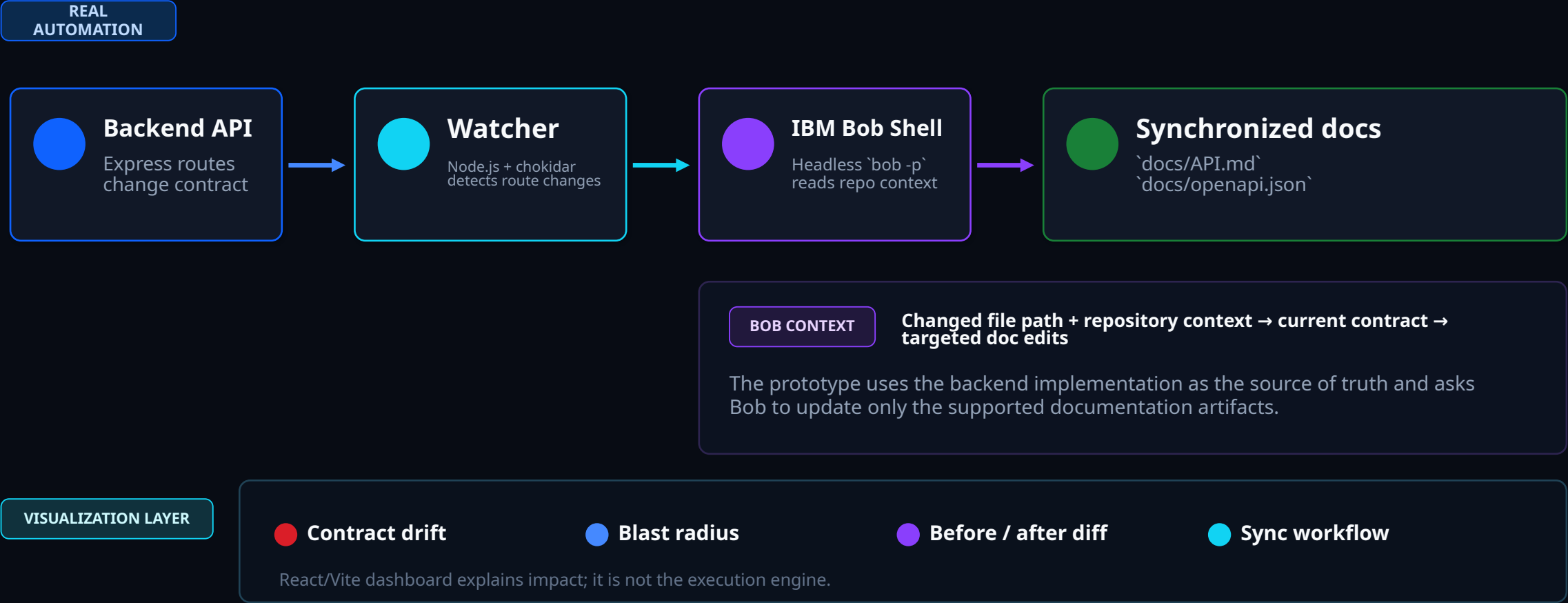
Teams coordinate manually

Developers hunt through files, tickets, and reviews to repair drift.

BridgeForge targets the gap between “code changed” and “everything downstream caught up.”

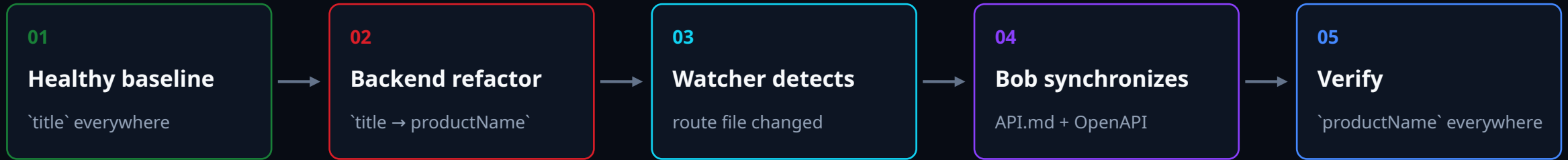
Turn IBM Bob into an automated synchronization layer

BridgeForge monitors backend changes, invokes Bob Shell, and updates dependent API documentation from the source of truth.



One rename. Five visible stages.

The entire prototype is built around a repeatable ``title` → productName`` API-contract change.



Backend contract



BEFORE

`"title": "Widget A"`

AFTER

`"productName": "Widget A"`

Bob-updated artifacts

docs/API.md

Human-readable tables + examples

productName

docs/openapi.json

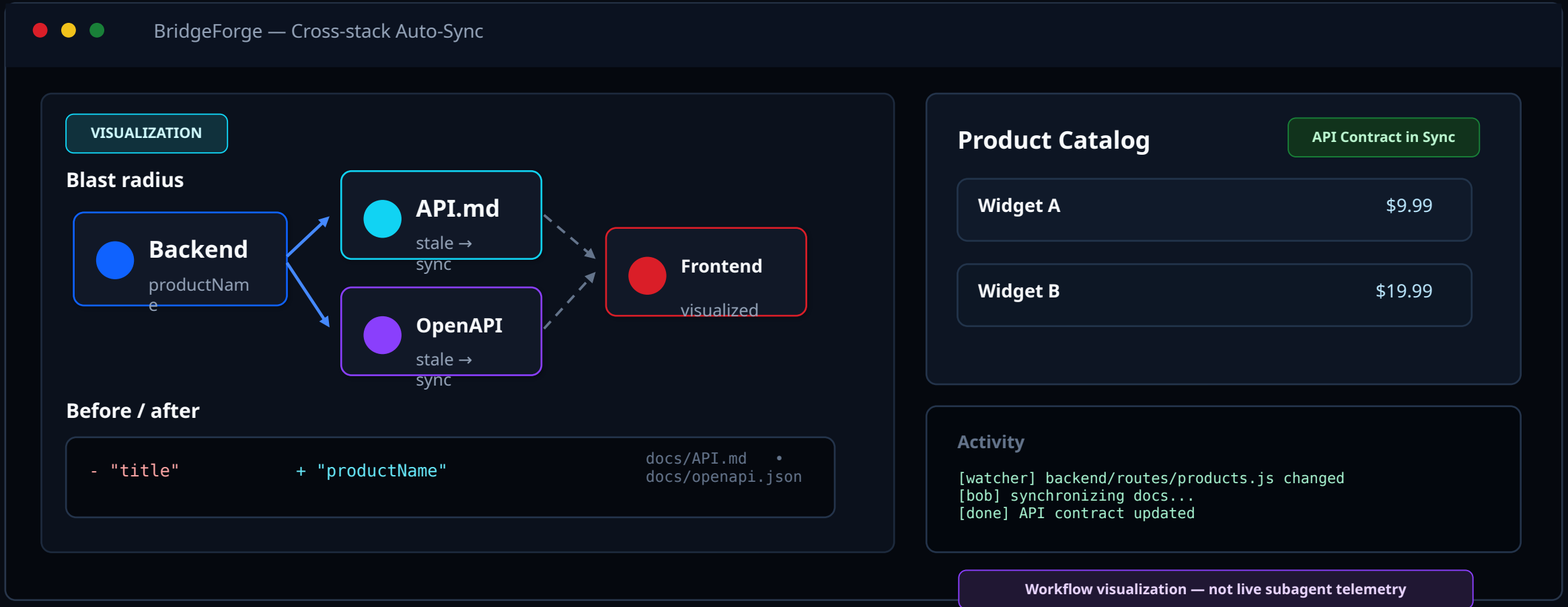
Machine-readable schema

productName

Source of truth restored across supported docs

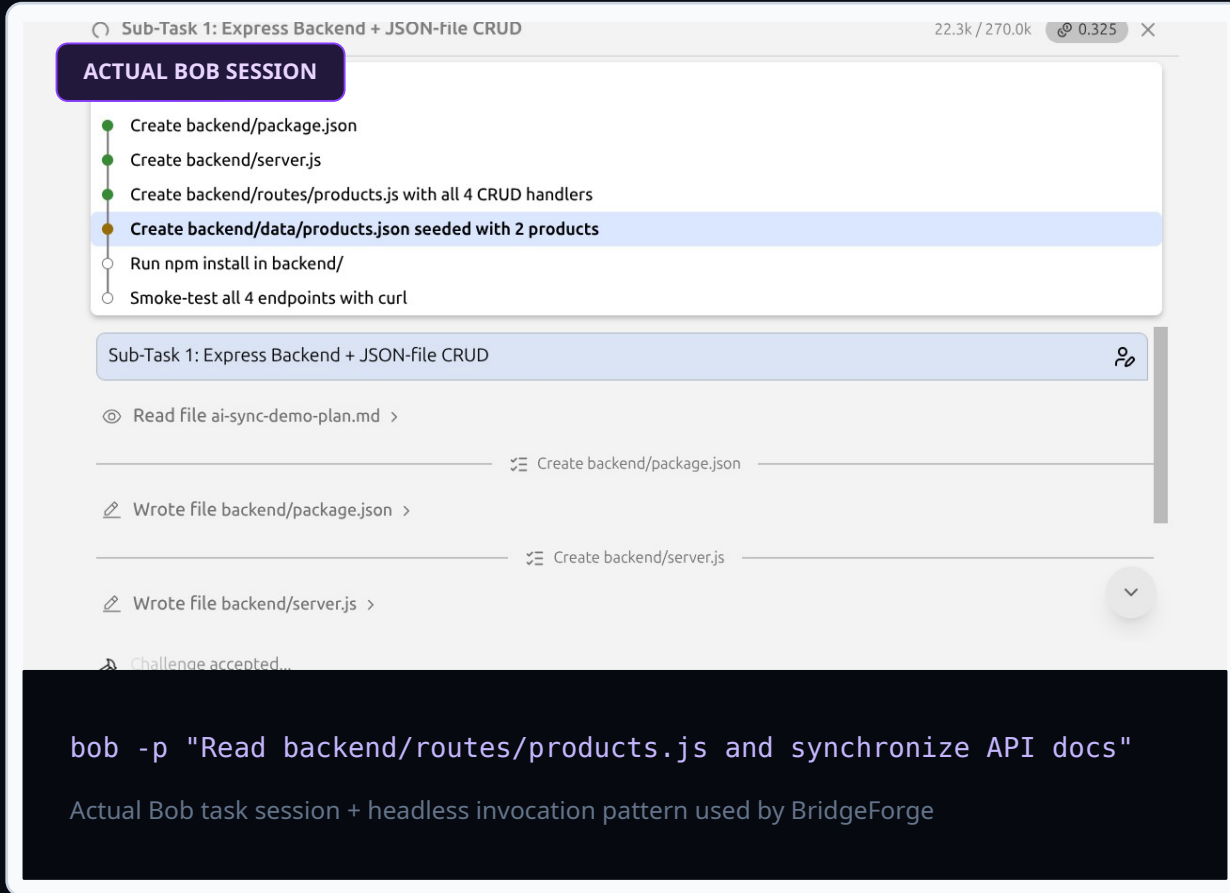
Make contract drift understandable, not invisible

The dashboard turns a backend mismatch into a visual story: what changed, what is affected, and what synchronization looks like.



From interactive assistant to automation primitive

BridgeForge invokes Bob programmatically and gives it repository context so synchronization can start when the code changes — not only when a developer asks.



The screenshot displays the Bob interface within a window titled 'Sub-Task 1: Express Backend + JSON-file CRUD'. At the top, a purple badge reads 'ACTUAL BOB SESSION'. Below it, a list of tasks is shown with green dots indicating completion: 'Create backend/package.json', 'Create backend/server.js', 'Create backend/routes/products.js with all 4 CRUD handlers', 'Create backend/data/products.json seeded with 2 products' (highlighted in blue), 'Run npm install in backend/', and 'Smoke-test all 4 endpoints with curl'. A task bar below the list shows 'Sub-Task 1: Express Backend + JSON-file CRUD' with a person icon. The main area shows a sequence of actions: 'Read file ai-sync-demo-plan.md >', 'Create backend/package.json' (with a checkmark icon), 'Wrote file backend/package.json >', 'Create backend/server.js' (with a checkmark icon), and 'Wrote file backend/server.js >'. A 'Challenge accepted...' message is visible at the bottom. A dark blue box at the bottom contains the command: `bob -p "Read backend/routes/products.js and synchronize API docs"` and the text: 'Actual Bob task session + headless invocation pattern used by BridgeForge'.

REAL TODAY

- Repository context** Bob reads the changed backend source and relevant project files.
- Headless invocation** The watcher calls a single ``bob -p`` task automatically.
- Concrete edits** Bob updates ``docs/API.md`` and ``docs/openapi.json``.

The key shift Change → detection → Bob → synchronized artifacts

No manual chat prompt is required to start the workflow.

Docs today. Cross-stack synchronization tomorrow.

The prototype proves the trigger → context → AI edit loop; the same pattern can expand to more downstream consumers.

PROTOTYPE NOW

Automated

- Detect backend route changes
- Invoke IBM Bob Shell
- Update API.md
- Update OpenAPI schema

NEXT

Extend the blast radius

- Frontend API bindings
- Tests + fixtures
- Generated SDKs
- Validation schemas

VISION

Cross-team auto-sync

- Map dependencies across repos
- Delegate specialized repair tasks
- Validate before applying changes
- Keep contracts continuously aligned

BridgeForge makes contract drift visible — and starts fixing it automatically.

IBM Bob as the synchronization layer between code changes and dependent engineering artifacts.